

Block Bootstrap Methods and the Choice of Stocks for the Long Run

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Financial advisors commonly recommend that the investment horizon should be rather long in order to benefit from the “time diversification”. In this case, in order to choose the optimal portfolio, it is necessary to estimate the risk and reward of several alternative portfolios over a long-run given a sample of observations over a short-run. Two interrelated obstacles in these estimations are lack of sufficient data and the uncertainty in the nature of the return generating process. To overcome these obstacles researchers rely heavily on block bootstrap methods. In this paper we demonstrate that the estimates provided by a block bootstrap method are generally biased and we propose two methods of bias reduction. We show that an improper use of a block bootstrap method usually causes underestimation of the risk of a portfolio whose returns are independent over time and overestimation of the risk of a portfolio whose returns are mean-reverting.

Keywords: Long-run; Time-series data; Serial dependence; Parameter estimation; Bootstrap; Block bootstrap

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1. Introduction

An important issue in finance is the relationship between the optimal portfolio choice and the length of the investment horizon. Despite the abundance of academic models that describe the investor's preferences, in theory the choice of the investor's optimal portfolio is pretty simple: select a portfolio that either maximizes the investor's expected utility or the investor's portfolio performance measure over a specific investment horizon. Yet the practical implementation of a theoretical model requires the estimation of the model's input parameters. This requires, among other things, estimating the risk and reward of each portfolio. Financial advisors and mutual fund sales literature commonly recommend that the investment horizon should be rather long in order to benefit from the "time diversification". In finance, the problem of estimating risk and reward over a long horizon comes from lack of sufficient data and uncertainty in the nature of the return generating process.

In particular, the majority of historical databases used in financial research have a span of about 80 years of monthly observations. The estimation of risk and reward of a stock portfolio on horizon of, say, 5 years using only 16 available non-overlapping 5-year return observations is subject to sizeable errors, to say nothing of errors of estimations on horizons of 15-20 years. To tackle this problem researchers typically rely on statistical bootstrap methods. A bootstrap is a computer-intensive method of estimation of parameters and distributions by resampling the original data. The practical realization of a bootstrap method, however, depends crucially on whether the returns are serially dependent or not. If the returns are independent over time, then one can employ the standard bootstrap method, suggested by Efron (1979), which is relatively straightforward in implementation. If the returns are serially dependent, then the standard bootstrap method cannot be used, because this method destroys any serial dependence in data. In order to capture the dependence structure in the return series while estimating the parameter of interest of the return distribution on long horizons, many researchers rely on the moving block bootstrap methods that were introduced by Hall (1985) and Künsch (1989). These methods use overlapping blocks of data instead of individual observations to estimate the parameters and distributions.

Taking into account the growth in popularity of bootstrap methods in financial applications, in this paper we aim to achieve several goals. First, for the sake of self-completeness, we make a brief review of statistical bootstrap methods. Second, we demonstrate that the estimates provided by a block bootstrap method are generally biased. This can result, for example, in a severe underestimation or overestimation of a portfolio risk. Third, we propose two methods of bias reduction. Forth, we argue that the correct accounting for the bias is impossible without a detailed study of the nature of serial dependency in return series.

In order to make the exposition of the paper more illustrative and concrete, we consider a specific "case study". In particular, we consider a long-term investor with mean-variance preferences. The goal of this investor is to rank the following four portfolios according to their attractiveness: small stocks, large stocks, growth stocks, and value stocks. Let $(p_1, p_2, \dots, p_{n+1})$ be the relevant investor's observations of the natural log of a stock portfolio price over n equally spaced time intervals. Denote the one-period log return during the interval t , $1 \leq t \leq n$, by

$$x_t = p_{t+1} - p_t,$$

where p_t and p_{t+1} are the prices of the portfolio at the beginning and at the end, respectively, of interval t . The resulting sample of n return observations is $X = (x_1, x_2, \dots, x_n)$. The probability distribution of x_t is unknown for the investor. We assume that the investor's investment horizon

is m time intervals and denote by $x_{t,t+m}$ the log returns over m successive intervals

$$x_{t,t+m} = p_{t+m} - p_t.$$

Since the investor exhibits mean-variance preferences, in order to rank the portfolios the investor can employ the Sharpe ratio which is computed as

$$S(x_{t,t+m}) = \frac{E[x_{t,t+m}] - r}{\sqrt{\text{Var}(x_{t,t+m})}},$$

where $E[x_{t,t+m}]$ is the expected return, $\text{Var}(x_{t,t+m})$ is the variance of returns, and r is the risk-free rate of return. To compute the Sharpe ratio of each stock portfolio, the investor needs to estimate the expected return and variance of returns. The most important obstacle in these estimations is that there is only a few available non-overlapping intervals of length m in the total sample of n observation. How can the investor overcome this obstacle?

To the best of the authors' knowledge, so far there are two predominating approaches, used in the academic literature, to estimating the long-run parameters of return distributions sampled over short-run. The first most simplified approach is to assume that the stock portfolio returns are independent and identically distributed and employ the standard bootstrap method. This approach was used by, for example, Lloyd and Modani (1983), Lloyd and Haney (1985), Leibowitz and Langetieg (1989), Butler and Domian (1991), Hodges *et al.* (1997), Hickman *et al.* (2001), Mukherji (2003), Sinha and Sun (2005), and Dierkes *et al.* (2010).

However, most practitioners believe that stock returns are mean-reverting. This means that stocks become less risky as the investment horizon increases. There is a large strand of academic literature which supports the practitioners' point of view (to mention just a few, see Summers (1986), Fama and French (1988), Poterba and Summers (1988), Lo and MacKinlay (1988), Campbell and Shiller (1998), Balvers *et al.* (2000), and Gropp (2004)). Among other things, this literature documents long-term reversal in the prices of some stock portfolios. As a response, the second seemingly more elaborate approach to estimating the long-run parameters of return distributions is to assume that there is some type of dependency in the return series and apply the moving block bootstrap method to all return series. Implicitly, one assumes that even though there might be no dependency in some return series, the moving block bootstrap method cannot produce worse estimates than the standard bootstrap method. Since the implementation of a moving block bootstrap method requires choosing a block length, in the literature there are two variants of this approach. In the first variant one chooses an arbitrary¹ fixed block length regardless of the horizon length (see Hansson and Persson (2000), Graffund (2001), Sanfilippo (2003), Jan and Wu (2008), and Mukherji (2011)). In the second variant one chooses the block length equal to the horizon length (see Lin and Chou (2003) and Beach (2007)). Surprisingly, the choices of the block length have never been properly justified and the consequences of using different block lengths have never been analyzed. One of the main purposes of this paper is to fill these voids.

The paper's organization is as follows. In Section 2 we review the statistical bootstrap methods, both for independent and serially dependent data, and point out that the statistical bootstrap methods were not developed, in the first place, to estimate the long-run parameters of processes sampled over the short-run. Moreover, the implementation of the moving block bootstrap method requires choosing the optimal block length. How to do it in the context of the financial problem

¹The usual justification for some particular block length is to say that "it is probably long enough to pick up most of possible time dependencies" (Hansson and Persson (2000), page 57).

considered in this paper still remains unanswered. In Section 3 we study whether the logs of the stock portfolio prices follow a random walk or there is some time dependency in the stock prices. For this purpose we perform statistical randomization tests where the null hypothesis is that the stock returns are independent over time. We find a strong statistical evidence of mean reverting behavior in the prices of small and value stocks.² In contrast, neither the large stocks nor the growth stocks present any evidence of mean reversion. Consequently, whereas the standard bootstrap can be applied to the estimation of the long-run parameters of the return distributions of the large and growth stocks, the moving block bootstrap is necessary in order to capture the dependence structure in the return series of small and value stocks.

In Section 4 we, first of all, present a well-established and flexible parametric model for the stock price process. In this model the log stock prices are composed of two components: a permanent component which is a random walk, and a transitory component which is an autoregressive process. Such a mixture of two components is able to generate long-term reversal in stock prices as documented in numerous empirical studies. We demonstrate that in this model a moving block bootstrap method generally produces biased estimates when either the block length is fixed irrespective the horizon length or the block length equals the horizon length. The bias is especially large when either the stock price process is a pure random walk and the block length is large, or the transitory component in the stock price dominates the permanent component and the block length is small. We show that the estimation bias can be mitigated by either the standard bias correction method or by choosing a proper block length.

In Section 5 we demonstrate that different choices for the block length in the moving block bootstrap method generally lead to different ranking of stock portfolios. In particular, we show that the use of a non-optimal block length usually causes underestimation of the risk of a portfolio whose returns are independent over time and overestimation of the risk of a portfolio whose returns are mean-reverting. Section 6 summarizes and concludes the paper. In the same section we provide readers with some recommendations related to the choice of a moving block bootstrap method and the block length.

2. Review of Bootstrap Methods

2.1. Setup

Let $X = (x_1, x_2, \dots, x_n)$ be observations from a strictly stationary time series $\{x_t\}$ (note that x_t can also be a vector). The assumption of stationarity implies that the joint probability law of $(x_t, x_{t+1}, \dots, x_{t+k})$ does not depend on t for any $k \geq 0$. We denote by F the probability distribution of x_t and assume that F is unknown. Let $\theta(F)$ be some parameter of interest such as the mean, median, correlation, or standard deviation of F . Let $\hat{\theta}(X)$ be an estimator of $\theta(F)$ computed using observations X .

Since the estimator $\hat{\theta}(X)$ is a random variable, for the purpose of statistical inference one needs to know the probability distribution of $\hat{\theta}(X)$. This is necessary for estimating standard errors and confidence intervals for $\theta(F)$. This is also necessary for constructing hypothesis tests about the value of $\theta(F)$. The bootstrap is a method for estimating the distribution of an estimator or test statistic by resampling the data. Bootstrapping is becoming the most popular general purpose approach to statistical inference because it does not require any parametric assumption on F , and because it can be effectively utilized with smaller sample sizes ($n < 30$). Bootstrapping is

²To the best of the authors's knowledge, we are the first to present a statistically significant evidence of mean reversion in the prices of portfolios of value stocks.

also often used as an alternative to inference based on parametric assumptions when theoretical calculation of the statistics of interest is not possible.

The practical realization of a bootstrap method depends crucially on whether the observations of $\{x_t\}$ are assumed to be independent or dependent. We will refer to the bootstrap method for independent data as the standard bootstrap. The most popular bootstrap methods for dependent data are block bootstraps. The standard bootstrap and some often used block bootstrap methods are reviewed below. The readers should be aware, however, that the bootstrap methods for dependent data is a lively research area. For a more comprehensive review of different bootstrap methods, the interested readers can consult Gentle *et al.* (2004).

2.2. Standard Bootstrap

Efron (1979) was the first to introduce the standard bootstrap method. The method is implemented by resampling the data randomly with replacement. More formally, this method consists in drawing random resamples $X^* = (x_1^*, x_2^*, \dots, x_n^*)$ from $(x_1, x_2, \dots, x_n)$. Note that one does not get a permutation since the values of X^* are selected with replacement. By doing this several times and computing for each resample $\hat{\theta}(X^*)$, one obtains an approximate probability distribution of the estimator $\hat{\theta}(X)$. The number of resamples should be large and is mainly limited by available computing power and time.

2.3. Block Bootstrap

If the data are dependent, in order not to break up the dependence while performing a bootstrap method, Hall (1985) suggested to resample the data using blocks of data instead of individual observations. There are basically two different ways of proceeding, depending on whether the blocks are overlapping or non-overlapping. Both approaches were introduced by Hall (1985) in the context of spatial data. Carlstein (1986) proposed non-overlapping blocks for univariate time series data, whereas Künsch (1989) suggested overlapping blocks in the same setting. We will refer to the methods of Carlstein (1986) and Künsch (1989) as the non-overlapping block bootstrap method and the moving block bootstrap method respectively.

More formally, let b and l denote integers such that $n = bl$. In particular, b denotes the number of blocks and l denotes a block length. In the non-overlapping block bootstrap method X is divided among b disjoint blocks, where the k th block is given by $B_k = (x_{(k-1)l+1}, x_{(k-1)l+2}, \dots, x_{kl})$ for $1 \leq k \leq b$. In the moving block bootstrap method the number of overlapping blocks is given by $n-l+1$, where the k th block is given by $B_k = (x_k, x_{k+1}, \dots, x_{k+l-1})$ for $1 \leq k \leq n-l+1$. The block bootstrapping method consists in choosing blocks $B_1^*, B_2^*, \dots, B_b^*$ by resampling randomly with replacement from among available blocks $B_1, B_2, \dots, B_N$ where N is either b or $n-l+1$ depending on whether blocks are non-overlapping or overlapping. If $B_i^* = (x_{i1}^*, x_{i2}^*, \dots, x_{il}^*)$, then the bootstrap version of X is

$$X^* = (B_1^*, B_2^*, \dots, B_b^*) = (x_{11}^*, x_{12}^*, \dots, x_{1l}^*, x_{21}^*, x_{22}^*, \dots, x_{2l}^*, x_{b1}^*, x_{b2}^*, \dots, x_{bl}^*).$$

Again, as in the standard bootstrap method, the number of resamples should be large and in practice is limited by available computing power and time.

By construction, in the moving block bootstrap method the bootstrapped time series has a nonstationary (conditional) distribution. The resample becomes stationary if the block length is random. This version of the moving block bootstrap is called the stationary bootstrap and was introduced by Politis and Romano (1994). In particular, unlike the original moving block bootstrap method where the block length is fixed, in the stationary block bootstrap method the

length of block k , l_k , is generated from a geometric distribution with probability p . Thus, the average block length equals $\frac{1}{p}$. The k th block begins from a random index i which is generated from the discrete uniform distribution on $\{1, \dots, n\}$. Since a generated block length is not limited from above, $l_k \in [1, \infty)$, and the block can begin with observation x_n , the stationary bootstrap method “wraps” the data around in a “circle”, so that x_1 follows x_n and so on.

Another method that can potentially improve the performance of the moving block bootstrap was suggested by Carlstein *et al.* (1998). This method consists in aligning with higher likelihood those blocks that match at their ends. This is achieved by resampling the blocks according to a Markov chain whose transition probabilities depend on data. This method is usually called either the “matched block bootstrap”, or the “Markov bootstrap” (“Markovian bootstrap”). Graflund (2001) and Sanfilippo (2003) implement a modification of the same idea. Instead of matching blocks at their ends, these authors match two subsequent blocks of data using their standard deviations.

The question of paramount importance in the implementation of the block bootstrap method is how to choose the optimal block length l . The paper by Hall *et al.* (1995) addresses this issue. The authors find that the optimal block length depends very much on context. In particular, the optimal asymptotic formula for block length is $l \sim Cn^{\frac{1}{h}}$, where C is a constant and $h = 3, 4$, or 5 . For computing block bootstrap estimators of bias (of $\hat{\theta}(X)$) or variance (of x_t), $h = 3$. For computing block bootstrap estimators of one-sided and two-sided distribution functions (of $\hat{\theta}(X)$), $h = 4$ and 5 respectively.

2.4. Discussion

Unlike the existing and well-understood bootstrap methods in statistics that have been reviewed in this section, in the context of the financial problem presented in the introductory section the investor is not interested in knowing the probability distribution of the estimator $\hat{\theta}(X)$. Instead, the investor wants to estimate the parameter of interest $\theta(F_m)$, where F_m is the probability distribution of log return over m successive intervals. Apparently, the reviewed bootstrap methods were not supposed to handle this kind of problem in the first place. Indeed, the bootstrap methods were developed to estimate the probability distribution of the estimator $\hat{\theta}(X)$ of $\theta(F)$ where the estimator is computed by a standard method, not to estimate $\theta(F_m)$ by some original method.

3. Stock Market Prices do not Always Follow Random Walks

3.1. Literature Review

Until the late 1980s there was a widespread agreement among the academic community that stock prices follow a random walk. Indeed, a large body of empirical literature seemed to support this point of view (see Fama (1970) and Leroy (1982) for surveys). The efficient market hypothesis is strongly associated with the idea of a random walk in stock prices and loosely says that stock returns are unpredictable. However, during the late 1980s, there appeared a series of papers where the authors challenged the random walk hypothesis (see, for example, Summers (1986), Campbell and Mankiw (1987), Fama and French (1988), Lo and MacKinlay (1988), and Poterba and Summers (1988)). In particular, these authors considered the time series properties of stock returns at increasing horizons and found the indications of mean-reversion. For example, Fama and French (1988) discovered a substantial negative autocorrelation in multi-period returns over long horizons. Poterba and Summers (1988) found that stock returns exhibit positive

and statistically significant autocorrelations in returns over horizons shorter than one year and negative, though not statistically significant at conventional levels (1% or 5%), autocorrelations over longer horizons.

However, the conclusions reached in these earlier papers were strongly criticized on statistical grounds. For example, Kim *et al.* (1991) demonstrated that due to the small-sample bias the statistical significance of the test statistics in Fama and French (1988) and Poterba and Summers (1988) was overstated and the post-war return series do not show any evidence of mean reversion. Similarly, Richardson and Stock (1989) and Richardson (1993) showed that correcting for the small-sample bias may reverse the results obtained by Fama and French (1988) and Poterba and Summers (1988).

Apparently, the statistical power of earlier tests was insufficient to reject the random walk hypothesis. Jegadeesh (1991) suggested a new, more powerful test and detected statistically significant evidence of mean reversion in stock returns not only during the overall period, but also during the post-war period. In addition, Jegadeesh found evidence of mean reversion not only for the U.S. stock market, but also for the U.K. stock market. Later on based on a panel approach Balvers *et al.* (2000) found statistically significant evidence of mean-reverting behavior in many international stock markets. Thus, mean reversion in stock returns seems to be an international phenomenon. Using the same technique as in Balvers *et al.* (2000), Gropp (2003) and Gropp (2004) found statistically significant evidence of mean reversion in the returns on the portfolios of small size stocks (also during the post-war period) and the industry-sorted portfolios. Moreover, Balvers *et al.* (2000), Gropp (2003), and Gropp (2004) showed that parametric contrarian investment strategies that exploit mean reversion outperform buy-and-hold and standard contrarian strategies. This provides further support for the mean reversion findings in these papers. Thus, nowadays the evidence of mean reversion in the returns on some stock portfolios seems to be manifested. In the next subsection we provide some new evidence on mean reversion in stock prices.

3.2. New Evidence on Mean Reversion in Stock Prices

3.2.1. Research Design

The question we want to study in this section is whether the log of a stock (portfolio) price follows a random walk or there is some time dependency in the stock prices. To answer this question we perform a well-known test based on the examination of the variance ratio. This test is very popular and was used, among others, by Cochrane (1988), Lo and MacKinlay (1988), and Poterba and Summers (1988). The test is based on the computation of the following test statistics

$$VR(m) = \frac{Var(x_{t,t+m})}{mVar(x_t)}. \quad (1)$$

The test is motivated by the notion that if the stock returns are independent and identically distributed (i.i.d.) random variables, then the variance ratio is unity irrespective of the period length m . In other words, without time dependency in data, the variance of m -period returns equals m times the variance of one-period returns. The null hypothesis of a random walk is rejected if the variance ratio is significantly different from unity.

We want to compute the variance ratio $VR(m)$ for return horizons from 2 to 20 years. The fundamental problem with these computations is that the majority of historical price data cover a span of no more than 80 years. Therefore in the computation of the test statistics we employ overlapping intervals. In particular, given a sample of n one-period return observations, the

number of overlapping m -period returns is $n - m + 1$ such that

$$x_{t,t+m} = \sum_{i=1}^m x_{t+i-1}.$$

The variance of m -period returns is computed as

$$\text{Var}(x_{t,t+m}) = E \left[(x_{t,t+m} - E[x_{t,t+m}])^2 \right],$$

for $1 \leq t \leq n - m + 1$. Unfortunately, the use of overlapping returns leads to some potentially very serious econometric issues which are commonly termed as “small-sample bias”. In particular, the estimate for the variance of multi-period returns, $\text{Var}(x_{t,t+m})$, is downward biased when one uses overlapping blocks of data.¹ As an immediate consequence, the estimate for the variance ratio $VR(m)$ becomes also downward biased. In addition, since the estimate for $VR(m)$ is a random variable, for the purpose of statistical inference we need to know the probability distribution of $VR(m)$. This is necessary for estimating standard errors and confidence intervals for $VR(m)$. This is also necessary for performing hypothesis tests about the value of $VR(m)$.

When the nature of the data generating process is completely unknown, it is generally not possible to tackle the econometric problems described above. However, in the context of the null hypothesis our goal is primarily to test whether stock returns are distributed independently of their ordering in time. Since under the null there is no dependency in return series, in order to estimate the significance level of the test statistics, we follow closely Kim *et al.* (1991) and Nelson and Kim (1993) where the authors employ the randomization method. The randomization method was introduced by Fisher (1935) and provides a very general and robust approach for computing the probability of obtaining some specific value for an estimator under the null hypothesis of no dependence. We refer the interested readers to Noreen (1989) and Manly (1997) for extensive discussion of the randomization tests. In a nutshell, randomization consists of reshuffling the data to destroy any dependency and then recalculating the test statistics for each reshuffling in order to estimate its distribution under the null hypothesis of no dependence. The great advantage of the randomization method is that it is very simple and no assumptions are made about the actual distribution of stock returns.

Observe that there is a great deal of similarity between the randomization method and the standard bootstrap method of Efron (1979). Whereas the standard bootstrap method is implemented by resampling the data randomly *with* replacement, the randomization method is implemented by resampling the data randomly *without* replacement. The difference between the randomization and bootstrap is in their justification. As Manly (1997) writes:

Randomization is based on the argument that the process producing the observed data is equally likely to produce them in any order, and it is of interest to see what proportion of the possible orders display a particular type of pattern. On the other hand, bootstrapping is justified by regarding the observed distribution in a time series as the best estimate of the distribution of values produced by the generating mechanism. Bootstrapping is therefore an alternative to assuming some particular theoretical distribution for series values.

Consider the assessment of the significance level of the estimate for the variance ratio $VR(m)$. First, we use the original series $X = (x_1, x_2, \dots, x_n)$ to obtain the actual historical estimates

¹For the sake of motivation, consider what happens to the estimate for $\text{Var}(x_{t,t+m})$ when $m \rightarrow n$. Obviously in the limit, when the period length converges to the sample length, there is only one available block of data to estimate $\text{Var}(x_{t,t+m})$. Therefore, regardless of the nature of the data generating process, $\text{Var}(x_{t,t+m}) \rightarrow 0$ as $m \rightarrow n$.

Table 1. Descriptive statistics of the annual log returns on four portfolios sorted with respect to size (market equity) and book-to-market ratio. ρ_k denotes the autocorrelation between x_t and x_{t+k} . The Jarque-Bera test statistics is used to test the null hypothesis that the returns are normally distributed. The critical value of this test statistics is 5.99 at 5% level. The null hypothesis is rejected when the value of the Jarque-Bera test statistics exceeds the critical value. Bold text indicates values that are statistically significant at the 5% level.

Statistics	Portfolio			
	Small stocks	Large stocks	Growth stocks	Value stocks
Mean, %	10.82	9.55	8.81	13.08
Std. dev., %	25.09	16.54	18.20	20.08
Skewness	-0.22	-0.85	-0.67	-0.51
Kurtosis	2.95	3.97	3.29	3.19
Jarque-Bera	0.50	10.12	4.88	2.86
ρ_1	-0.01	0.03	-0.01	-0.13
ρ_2	-0.17	-0.16	-0.14	-0.26
ρ_3	0.01	0.04	-0.03	0.05
ρ_4	0.20	0.14	0.18	0.15
ρ_5	-0.26	-0.12	-0.13	-0.07
ρ_6	-0.20	0.02	0.01	-0.02
ρ_7	-0.17	0.12	0.08	-0.01
ρ_8	0.04	0.14	0.17	-0.03
ρ_9	-0.04	0.05	-0.01	0.21
ρ_{10}	-0.01	-0.14	-0.17	-0.13

for $VR(m)$. Then we randomize the original series to get a permutation $X^* = (x_1^*, x_2^*, \dots, x_n^*)$. This is repeated a large number of times, each time obtaining an estimate for $VR^*(m)$. In this manner we estimate the sampling distribution of $VR(m)$ under the null hypothesis. Finally, to estimate the significance level for some particular m , we count how many times the computed value for $VR^*(m)$ after randomization falls below the value of the actual historical estimate for $VR(m)$. In other words, under the null hypothesis we compute the probability of obtaining a more extreme value for the variance ratio than the actual historical estimate. Note that in this manner we compute p-values of one-tailed test. In the rest of the section we will report the randomization estimates for the variance ratios, with corresponding p-values, for some stock portfolios. In each case our estimates and p-values are based on 10,000 reshuffles.

3.2.2. Mean Reversion in the Prices of Portfolios Formed on Size and Book-to-Market

The data for this empirical study are annual log (nominal) returns on four U.S. value-weighted portfolios formed on size (market equity) and book-to-market ratio: small stocks, large stocks, growth stocks, and value stocks. The data covers the post-war period from 1946 to 2009 (63 annual observations) and are obtained from the data library of Kenneth French.¹ In particular, the small and large stock portfolios are the bottom 30% and top 30%, respectively, of all NYSE, AMEX, and NASDAQ stocks sorted according to size. Similarly, the growth and value stock portfolios are the bottom 30% and top 30%, respectively, of all NYSE, AMEX, and NASDAQ stocks sorted according to book-to-market ratio. Table 1 presents the descriptive statistics for the returns on these four portfolios. Observe that all portfolios exhibit negative skewness and positive excess kurtosis (except the small stocks). Yet, on the basis of the Jarque-Bera test statistics, the assumption of normally distributed returns can be rejected for only large stocks. Nevertheless, the small stocks have a statistically significant negative autocorrelation at lag 5, and the value stocks have a statistically significant negative autocorrelation at lag 2. This already indicates that there is some dependency in the return series of the small and value stocks.

¹See http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Table 2. Variance ratios of the multi-period returns on four portfolios sorted with respect to size (market equity) and book-to-market ratio. Variance ratios are not corrected for the bias. The values in the brackets report the p-values of one-tailed test for the null hypothesis of no dependence. These p-values are obtained using the randomization method. Bold text indicates values that are statistically significant at the 5% level.

Horizon, years	Portfolio			
	Small stocks	Large stocks	Growth stocks	Value stocks
2	0.94 (0.38)	0.97 (0.48)	0.94 (0.36)	0.84 (0.14)
3	0.79 (0.17)	0.83 (0.25)	0.80 (0.20)	0.58 (0.01)
4	0.71 (0.14)	0.78 (0.24)	0.73 (0.18)	0.44 (0.00)
5	0.74 (0.24)	0.81 (0.36)	0.77 (0.30)	0.44 (0.01)
6	0.69 (0.23)	0.80 (0.39)	0.75 (0.33)	0.43 (0.02)
7	0.61 (0.19)	0.81 (0.45)	0.75 (0.37)	0.41 (0.03)
8	0.52 (0.14)	0.84 (0.53)	0.77 (0.44)	0.41 (0.05)
9	0.45 (0.11)	0.87 (0.59)	0.79 (0.50)	0.37 (0.04)
10	0.40 (0.09)	0.86 (0.61)	0.78 (0.51)	0.37 (0.06)
11	0.35 (0.07)	0.81 (0.57)	0.71 (0.47)	0.36 (0.07)
12	0.29 (0.05)	0.75 (0.54)	0.67 (0.45)	0.31 (0.06)
13	0.25 (0.03)	0.72 (0.54)	0.64 (0.45)	0.28 (0.05)
14	0.23 (0.03)	0.70 (0.54)	0.62 (0.46)	0.27 (0.06)
15	0.22 (0.04)	0.69 (0.56)	0.61 (0.48)	0.27 (0.08)
16	0.20 (0.04)	0.69 (0.58)	0.60 (0.50)	0.26 (0.08)
17	0.18 (0.04)	0.66 (0.58)	0.58 (0.51)	0.23 (0.07)
18	0.18 (0.05)	0.64 (0.59)	0.56 (0.52)	0.22 (0.08)
19	0.21 (0.09)	0.62 (0.60)	0.54 (0.52)	0.24 (0.12)
20	0.19 (0.09)	0.58 (0.58)	0.51 (0.51)	0.22 (0.12)

The results of the randomization tests are presented in Table 2. First of all, this table reports the variance ratios computed according to formula (1) using overlapping blocks of data. Apparently, for all the series the variance ratios decline with increasing horizon. Recall, however, that the estimates for the variance ratios presented in this table are downward biased under the null hypothesis of no dependence.² In order to find out whether the observed declines are statistically significantly different from the expected declines under the null hypothesis, this table reports the p-values of the test statistics under the null.

Observe that the small and value stock portfolios exhibit clear signs of mean reversion. In particular, for the small stocks the estimates for the variance ratios are statistically significantly below unity at 5% level at lags 12-18 years. At 10% level, we can reject the null hypothesis of no dependence at lags 10-20 years. Consequently, our findings agree with the results reported by Gropp (2003) (note, however, that we use another test statistics). For the value stocks the estimates for the variance ratios are statistically significantly below unity at 5% level at lags 3-9, and 13 years. At 10% level, we can reject the null hypothesis of no dependence at lags 3-18 years. Thus, we have a strong statistical evidence of mean reverting behavior in the prices of small and value stocks. In contrast, neither the large stock portfolio nor the growth stock portfolio present any evidence of mean reversion.

²We do not correct the estimates for the bias, because in order to properly correct for the bias we need to make some specific assumptions about the nature of the data generating process. This will be demonstrated later in the paper.

4. Implementation of the Block Bootstrap Method

4.1. Setting the Stage

If the one-period returns are independent and identically distributed, then it is appropriate to use the standard bootstrap method. If, on the other hand, there is some dependency in stock returns, the standard bootstrap method cannot be used. This is because the standard bootstrap destroys any serial dependence in data. In order to capture the dependence structure in the return series while estimating the parameter of interest of the return distribution on long horizons, many researchers rely on a block bootstrap method. Yet, as we already noted, the well-known statistical bootstrap methods reviewed in Section 2 were not supposed to handle this kind of problem in the first place. Moreover, recall that the implementation of the moving block bootstrap method requires choosing the optimal block length. How to do it in this context still remains unanswered, and the researchers have never properly justified the choice of the block length. In particular, the block length varies from paper to paper. The researchers employ either a fixed block length or a variable block length. In the former case the block length does not depend on the horizon length and varies from 6 months to 5 years. In the latter case the block length equals the horizon length.

The objectives of this section are two-fold. First, we demonstrate that the estimates provided by a block bootstrap method are generally biased. Second, we demonstrate how the bias can be corrected by estimating the magnitude of the bias. We also present the other alternative, namely, how to choose a proper block length to mitigate the bias. It turns out that the correct accounting for the bias seems to be impossible without making some specific assumptions about the nature of the return generating process. Therefore, in order to demonstrate the bias and the methods of bias reduction, first of all in this subsection we construct an approximate but flexible parametric model for the log stock portfolio price. The main criterion for the choice of a model is its ability to reproduce the observed empirical phenomena, in particular, a decreasing variance ratio.

After having studied a number of possible parametric specifications, we decided to employ a well-established model for the stock portfolio price. This model was for the first time considered by Muth (1960). Later on, this model was used in many papers and books, among others by Summers (1986), Campbell and Mankiw (1987), Fama and French (1988), Lo and MacKinlay (1988), Poterba and Summers (1988), and Campbell *et al.* (1997). In this model the log of a stock portfolio price is composed of two components: a *permanent* component which is a random walk, and a *transitory* component which is a stationary process

$$p_t = q_t + z_t. \quad (2)$$

In particular, the process q_t is a random walk

$$q_t = q_{t-1} + \mu + e_t, \quad (3)$$

where μ is expected drift and e_t is white noise. z_t is any zero-mean stationary process. The common interpretation is that q_t is the “fundamental” component that reflects the efficient market value, and z_t reflects a deviation from the efficient market value q_t , implying the presence of “fads” or “bubbles”.

To make the model as simple as possible, similarly to Fama and French (1988) we assume that z_t is an Autoregressive process of order 1 (AR(1))

$$z_t = \phi z_{t-1} + w_t, \quad (4)$$

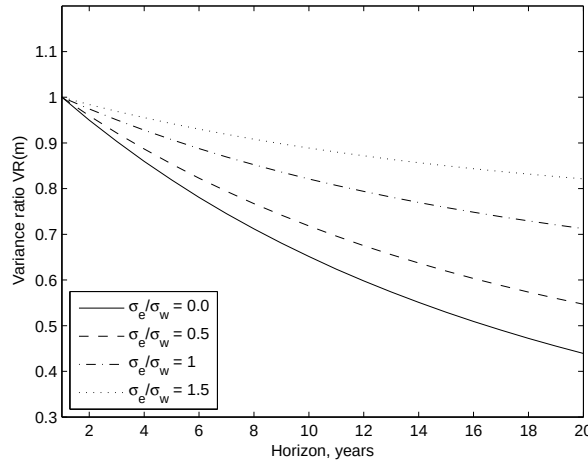


Figure 1. Theoretical variance ratio for $\phi = 0.9$ and different values of the ratio $\frac{\sigma_e}{\sigma_w}$.

where w_t is white noise and ϕ is a constant. e_t and w_t are assumed to be independent, that is, $Cov(e_t, w_t) = 0$. The return from t to $t + m$, $m \geq 1$, is given by

$$x_{t,t+m} = p_{t+m} - p_t = (q_{t+m} - q_t) + (z_{t+m} - z_t).$$

If in this model ϕ is close to unity,¹ then it is very difficult to detect the presence of the autoregressive component by studying the time series properties of one-period returns. Yet, this component might be detected by studying the time-series properties of multi-period returns. For example, one can examine the variance ratio. Using the standard logic, the derivation of the expression for $VR(m)$ for our model given by (2)-(4) yields (see Appendix for the details of the derivation)

$$VR(m) = \frac{m(1 - \phi^2) \left(\frac{\sigma_e}{\sigma_w} \right)^2 + 2(1 - \phi^m)}{m(1 - \phi^2) \left(\frac{\sigma_e}{\sigma_w} \right)^2 + 2m(1 - \phi)}, \quad (5)$$

where σ_e and σ_w are the standard deviations of e_t and w_t respectively.

Figure 1 demonstrates possible shapes of the $VR(m)$ for $\phi = 0.9$ and different values of the ratio $\frac{\sigma_e}{\sigma_w}$. Observe that whereas the random walk component produces white noise in returns, the stationary component causes a decrease in the variance ratio on longer horizons. Thus, the presence of the AR(1) component can generate long-term reversal in stock prices documented by numerous empirical studies. Finally note that the smaller the value of the ratio $\frac{\sigma_e}{\sigma_w}$, the more pronounced the effects of the AR(1) component.

Before proceeding to the bias demonstration and the methods of bias reduction, we need to choose a range of feasible values for the ratio $\frac{\sigma_e}{\sigma_w}$. Fortunately, the estimates for $\frac{\sigma_e}{\sigma_w}$ are readily available in the literature. According to Poterba and Summers (1988), for the value-weighted

¹The reasonable value of ϕ should be close to but less than 1.0, see the discussion in Fama and French (1988).

U.S. stock market index the range of possible values for this ratio is $\frac{\sigma_e}{\sigma_w} \in [0.9, 1.3]$. For the equal-weighted U.S. stock market index the range of possible values for this ratio is $\frac{\sigma_e}{\sigma_w} \in [0.3, 0.7]$. Note that the equal-weighted index shows a greater degree of mean reversion than the value-weighted index. Since the equal-weighted index is biased towards small cap stocks, the findings of Poterba and Summers (1988) can be interpreted as it is basically the small cap stocks that exhibit mean reverting behavior, but not the large-cap stocks. Similarly, Lo and MacKinlay (1989) estimate the range of possible values to be $\frac{\sigma_e}{\sigma_w} \in [0.7, 1.4]$, which is close to the estimates provided by Poterba and Summers (1988). Yet, in the seminal paper by Shiller (1981), the author argues that the stock prices are too volatile, that is, the movements in stock price indexes could not realistically be attributed to changes in fundamental values. In his view, $\sigma_e \ll \sigma_w$, which states that the variance of the fundamental (permanent) component is much less than the variance of the transitory component. Using a simple efficient market model, Shiller shows that for a broad stock market index $\frac{\sigma_e}{\sigma_w} \in [0.1, 0.2]$. The big discrepancy between the estimates provided by Poterba-Summers and Lo-MacKinlay from one side, and Shiller from the other side, stems from the use of different estimation methods and the need of making some assumptions in the process of estimation. Since the researchers do not agree on the value of the ratio $\frac{\sigma_e}{\sigma_w}$, in our study we consider a wider range of possible values of this ratio, in particular we assume that $\frac{\sigma_e}{\sigma_w} \in [0.0, 1.5]$.

4.2. Demonstration of the Bias

Observe that in the model for the stock price process, presented in the previous subsection, the AR(1) component is transitory, but its effects are highly persistent through time. That is, the effects of the AR(1) component do not disappear after some time, but persist on all lags and horizons. Given this knowledge it is tempting to conclude that in this model the optimal block length in a block bootstrap method should equal the horizon length. However, this is not so because the estimations of the model parameters using overlapping blocks of data are subject to the same “small-sample bias” that was described in Section 3.2.1.

The demonstration of the bias in the estimation of the variance ratio can be easily done using a simulation study. In particular, using our model we simulate 5,000 times a series of returns with the following model parameters: $n = 63$ and $\phi = 0.9$. The value of the ratio $\frac{\sigma_e}{\sigma_w}$ takes either 0.0 or 1.5. Note that the length of a series, n , equals the number of annual observations in our empirical sample of the stock portfolio returns (see Section 3.2.2). After each simulation we estimate $VR(m)$ using three different moving block bootstrap methods: the (original) moving block bootstrap method of Künsch (1989); the matched (moving) block bootstrap method of Sanfilippo (2003); and the stationary (moving) block bootstrap method of Politis and Romano (1994). In the moving block bootstrap and matched block bootstrap methods we set the block length equal to either the horizon length, $l = m$, or $l = 5$ years. In the stationary block bootstrap method we also set the average block length equal to either the horizon length or 5 years. In each method we set the number of resampled blocks to be equal to 1,000. At the end of the simulations we compute the mean values of estimated $VR(m)$ and compare them to the true values given by expression (5).

The results of our simulations and estimations for the case where the block length equals the horizon length are presented in Figure 2. Observe that for the moving block bootstrap method the estimation bias is always negative. In other words, the bias always decreases the estimates for variance as compared to its true values. We find that the magnitude of the estimation bias depends on the sample size, block length, and the model parameters, mainly on the value of the ratio $\frac{\sigma_e}{\sigma_w}$. In particular, the magnitude of the bias decreases when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ decreases (see a similar note in Fama and French (1988), page 253). That is, the larger

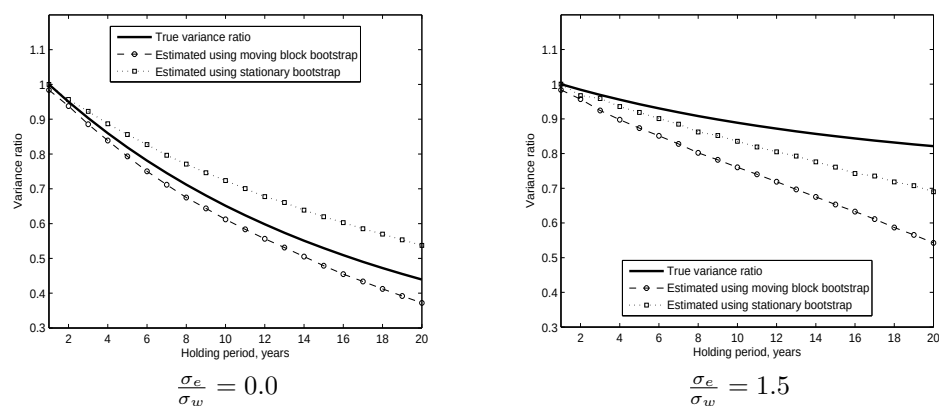


Figure 2. Demonstration of the bias in the estimations of the variance ratio using the *moving block bootstrap* and *stationary bootstrap* methods when the block length equals the horizon length.

the proportion of the variance of the transitory component in the total variance of the market returns, the smaller the estimation bias. Virtually similar results for the estimation bias are obtained for the matched block bootstrap method of Sanfilippo (2003). Yet, the matched block bootstrap method is much more time-consuming as one needs to compute a matching statistic and many generated blocks are rejected because of mismatching. For the stationary bootstrap method, the estimation bias can be both negative and positive. In particular, the bias is positive when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is rather low and negative when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is rather high.

The results of our simulations and estimations for the case where the block length equals 5 years are presented in Figure 3. Here note that with a fixed block length the estimation bias can be both negative and positive for each block bootstrap method. The bias can be very substantial when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is rather low and the horizon length is rather long. It is important to emphasize that in this case the use of the stationary bootstrap method provides a very smooth curve for the estimates of the variance ratio. In contrast, the use of the moving block bootstrap method of Künsch (1989) produces a curve that descends in a zigzag manner or pattern. This is a direct consequence of the non-stationarity of a pseudo-time series generated by the moving block bootstrap.¹

Finally in this subsection we would like to draw the reader's attention to the fact that a block bootstrap method with a fixed block length might be optimal for some specific horizon length. In other words, if a block bootstrap method is implemented with a fixed block length, then for some specific horizon length it produces an estimate for the variance ratio that is very close to the true value. For example, when $\frac{\sigma_e}{\sigma_w} = 0.0$ the 5-year block length is optimal when one uses the moving block bootstrap method and when the horizon length also equals 5 years. When one uses the stationary bootstrap method, on the other hand, the 5-year block length is optimal when the horizon length is about 2 years. When $\frac{\sigma_e}{\sigma_w} = 1.5$ the 5-year block length is optimal when one uses the moving block bootstrap method and when the horizon length is about 7-11 years. Similarly, when one uses the stationary bootstrap method, the 5-year block length is optimal when the horizon length is about 16-20 years. See Figure 3 for the illustrations.

¹See also the results of the simulation analysis presented by Politis and Romano (1994). These authors also demonstrate that the estimates provided by the stationary bootstrap are much less variable than those provided by the moving block bootstrap.

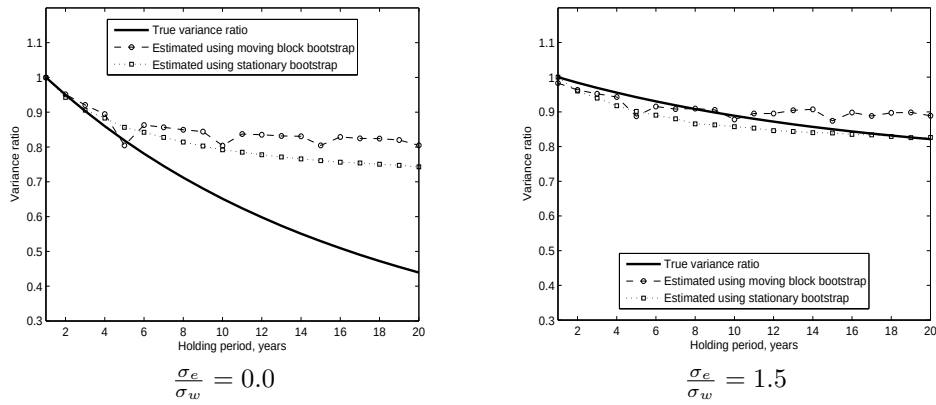


Figure 3. Demonstration of the bias in the estimations of the variance ratio using the *moving block bootstrap* and *stationary bootstrap* methods when the block length equals 5 years.

4.3. A Simple Bias Correction Method

Denote by $\widehat{Var}(x_{t,t+m})$ the estimator of $Var(x_{t,t+m})$ computed using observations X . We know that our estimator is generally biased, therefore, the estimator must be corrected for the bias. The estimation bias is defined as the difference between the expected and the true value of the estimator, that is, $Bias[\widehat{Var}(x_{t,t+m})] = E[\widehat{Var}(x_{t,t+m})] - Var(x_{t,t+m})$. The standard bias correction method consists in subtracting the bias from the estimator, that is, the bias adjusted values are computed as $\widehat{Var}(x_{t,t+m}) - Bias[\widehat{Var}(x_{t,t+m})]$. The bias can be estimated using a simulation method. In particular, one needs to choose some specific parameters of the model, choose an arbitrary block length l , simulate the return series many times, estimate $E[\widehat{Var}(x_{t,t+m})]$ for each horizon m , and finally find the bias.

For practical purposes it makes sense to assess the bias in the estimation of the variance ratio, because in this case the bias does not depend on the specific values of σ_e and σ_w , but only on the ratio $\frac{\sigma_e}{\sigma_w}$. When one knows the bias in the estimation of the variance ratio, the bias adjusted variance can be computed as

$$\widehat{Var}(x_{t,t+m}) - Bias[\widehat{VR}(m)] m \widehat{Var}(x_t),$$

where $Bias[\widehat{VR}(m)]$ is the bias in the estimation of the $VR(m)$, and $\widehat{Var}(x_{t,t+m})$ and $\widehat{Var}(x_t)$ are the estimated variances of m -period and one-period returns respectively. One needs to remember to estimate $\widehat{Var}(x_{t,t+m})$ using the same block length as in the estimation of $Bias[\widehat{VR}(m)]$.

Also for practical purposes it makes sense to use the block length that is equal to the horizon length, $l = m$. In this particular case the moving block bootstrapping becomes completely unnecessary. This is because in this case the probability distribution of F_m is implicitly approximated by the sample probability distribution $\hat{F}_m$ where the number of observations equals the number of overlapped blocks, $n - m + 1$, and the probability mass of each block equals $\frac{1}{n-m+1}$. Thus, as the number of simulations goes to infinity, the estimation of a parameter of interest of F_m converges to the computation of the value of this parameter using the discrete probability distribution $\hat{F}_m$. As a result, the estimation of a parameter of interest and the bias correction take virtually no time.

Table 3 reports the magnitudes of the bias in the estimation of the variance ratio using either the moving block bootstrap method or the stationary block bootstrap method. The bias is

Table 3. Bias in the estimation of the variance ratio, $E[\widehat{VR}(m)] - VR(m)$, where $VR(m)$ is the true variance ratio, given $\phi = 0.9$, the length of the investment horizon $m \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$, the value of the ratio $\frac{\sigma_e}{\sigma_w} \in \{0.0, 0.5, 1.0, 1.5\}$, and sample length $n \in \{63, 120\}$. In the moving block bootstrap the block length equals the horizon length. Similarly, in the stationary bootstrap the average block length equals the horizon length.

Sample length	Ratio σ_e/σ_w	Horizon, years									
		2	4	6	8	10	12	14	16	18	20
Panel A: Moving Block Bootstrap											
63	0.0	-0.00	-0.00	-0.00	-0.00	-0.01	-0.02	-0.02	-0.03	-0.04	-0.04
	0.5	-0.00	-0.02	-0.02	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09	-0.11
	1.0	-0.01	-0.02	-0.04	-0.07	-0.09	-0.11	-0.14	-0.16	-0.19	-0.21
	1.5	-0.01	-0.03	-0.06	-0.08	-0.11	-0.14	-0.17	-0.20	-0.23	-0.26
120	0.0	-0.00	-0.00	-0.00	-0.00	-0.00	-0.01	-0.01	-0.01	-0.01	-0.01
	0.5	-0.00	-0.00	-0.00	-0.01	-0.01	-0.02	-0.02	-0.03	-0.04	-0.04
	1.0	-0.00	-0.01	-0.02	-0.03	-0.04	-0.05	-0.06	-0.07	-0.08	-0.09
	1.5	-0.00	-0.02	-0.03	-0.05	-0.06	-0.07	-0.09	-0.10	-0.11	-0.13
Panel B: Stationary Bootstrap											
63	0.0	0.01	0.03	0.05	0.06	0.07	0.07	0.08	0.09	0.09	0.09
	0.5	0.00	0.02	0.03	0.04	0.04	0.04	0.04	0.05	0.04	0.04
	1.0	-0.01	0.00	-0.00	-0.00	-0.01	-0.01	-0.02	-0.03	-0.04	-0.05
	1.5	-0.02	-0.02	-0.03	-0.05	-0.06	-0.08	-0.09	-0.10	-0.12	-0.13
120	0.0	0.02	0.04	0.06	0.07	0.09	0.09	0.10	0.10	0.11	0.11
	0.5	0.01	0.03	0.04	0.05	0.06	0.06	0.06	0.07	0.07	0.07
	1.0	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.00
	1.5	-0.00	-0.00	-0.01	-0.01	-0.02	-0.02	-0.03	-0.04	-0.05	-0.05

estimated for different horizons length, different values of the ratio $\frac{\sigma_e}{\sigma_w}$, and different sample lengths.¹ This table can be used for the bias correction. In addition, this table presents yet another convincing demonstration of the existence of the bias in the estimation of the variance ratio using a block bootstrap method.

4.4. The Choice of the Optimal Block Length

Why the bias in the moving block bootstrap method is always negative when the block length equals the horizon length? Why the bias in the same method can be both positive and negative when the block length is less than the horizon length? Intuitively, the bias occurs because the blocks are overlapped. The use of overlapping blocks of returns makes the multi-period returns to be negatively serially dependent even if there is no serial dependence in one-period returns.¹ If the returns exhibit some degree of negative serial correlation, then the use of overlapping blocks increases the degree of negative serial dependence. Now recall that if the returns are dependent and we use the standard bootstrap method, then this method completely destroys any dependence in returns. Similarly, if the returns are serially dependent and we use relatively short blocks in the moving block bootstrap method, then such a method partially destroys the serial dependence in returns. The shorter the blocks, the bigger the destruction. Consequently, when shorter blocks do not fully destroy the artificially induced (as a result of using overlapping

¹The first sample length in the table equals the length of our empirical data set, see Subsection 3.2.2. The second sample length is roughly two times the length of our data set. This choice is made to demonstrate that a longer sample length decreases the estimation bias.

¹See Orcutt and Irwin (1948) and Marriott and Pope (1954) who describe in details why the use of overlapping blocks of data induces negative serial dependence.

blocks) serial dependence, the bias remain negative. Alternatively, blocks which are too short destroy not only the artificially induced serial dependence, but also destroy the existing serial dependence, and the bias becomes positive.

Similarly we can also explain why the bias in the stationary bootstrap method can be both positive and negative (when the average block length equals the horizon length). First, note that when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is relatively small, the artificially induced serial dependence is also small. In contrast, when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is relatively high, the induced serial dependence is also high. Second, note that a random block length also partially destroys serial dependence in data. Yet, the degree of “destruction” is the same regardless of the magnitude of the bias. Thus, when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is relatively small, the stationary bootstrap decreases the serial dependence more than necessary to remove the bias, so the resulting bias becomes positive. On the other hand, when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ is relatively high, the stationary bootstrap decreases the serial dependence less than necessary to remove the bias, so the resulting bias remains negative.

This knowledge, about how the block length affects the estimation bias, gives rise to the idea that the estimation bias can be reduced by the proper choice of the block length (this idea is exploited in, for example, Hall *et al.* (1995)). One can employ a simulation method to find the optimal block length in a block bootstrap method. The criterion of optimality for the choice of the block length is the minimization of the estimation bias of the parameter of interest. Since one of our goals is to estimate the long-run variance of stock returns, the criterion of optimality might be to choose the block length to minimize the bias in the estimation of the variance ratio. The simulations proceed as follows. Using our model, we simulate 5,000 times a series of returns given a set of the model parameters: n , ϕ , and $\frac{\sigma_e}{\sigma_w}$. After each simulation we estimate $VR(m)$ using either the moving block bootstrap method or the stationary bootstrap method where the block length varies from one to some maximum length, $l \in [1, \dots, l_{Max}]$ and the number of resamples is set to be equal to 1,000. At the end of the simulations we compute the mean values of estimated $VR(m)$ for each block length and find the block length which produces the value of estimated mean variance ratio closest to the theoretical value given by expression (5). The results of this exercise are reported in Table 4.

The results suggest the following:

- Holding the horizon length constant, the optimal block length decreases when the sample length decreases;
- Holding the horizon length constant, the optimal block length decreases when the value of the ratio $\frac{\sigma_e}{\sigma_w}$ increases.

Observe that the optimal block length in the moving block bootstrap method is roughly equal to the horizon length when $\sigma_w \gg \sigma_e$ (that is, when the transitory component in the stock price dominates the permanent component). Generally, for the moving block bootstrap method, the optimal block length is shorter than the horizon length. Note also that as $\frac{\sigma_e}{\sigma_w} \rightarrow \infty$, that is, as the transitory component disappears, the optimal block length converges to unity. In other words, in this case the moving block bootstrap reduces to the standard bootstrap. For the stationary bootstrap method, on the other hand, the optimal block length can be both shorter and longer than the horizon length.

5. Ranking of Stock Portfolios over Alternative Investment Horizons

The goal of this section is to demonstrate that different choices for the block length in the moving block bootstrap method generally lead to different ranking of stock portfolios. In this

Table 4. Optimal block length (in years) for the estimation of the variance ratio, $VR(m)$, given $\phi = 0.9$, the length of the investment horizon $m \in \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$, the value of the ratio $\frac{\sigma_e}{\sigma_w} \in \{0.0, 0.5, 1.0, 1.5\}$, and sample length $n \in \{63, 120\}$. The block length in the moving block bootstrap and the average block length in the stationary bootstrap is limited from above to 60 years.

Sample length	Ratio σ_e/σ_w	Horizon, years									
		2	4	6	8	10	12	14	16	18	20
Panel A: Moving Block Bootstrap											
63	0.0	2	4	6	8	10	12	14	16	18	20
	0.5	2	4	6	8	10	11	13	15	17	19
	1.0	1	3	5	7	9	11	11	12	13	14
	1.5	1	2	3	4	5	7	7	8	8	9
120	0.0	2	4	6	8	10	12	14	16	18	20
	0.5	2	4	6	8	10	12	14	16	18	20
	1.0	2	4	6	7	9	11	13	15	17	19
	1.5	1	3	5	7	9	10	11	12	13	13
Panel B: Stationary Bootstrap											
63	0.0	6	14	16	28	36	44	56	60	60	60
	0.5	2	8	10	14	16	20	26	30	32	34
	1.0	2	4	6	6	8	8	8	10	10	12
	1.5	2	2	4	4	4	4	8	8	8	8
120	0.0	6	14	26	36	58	58	60	60	60	60
	0.5	14	14	24	32	38	44	44	50	50	50
	1.0	4	6	8	12	16	16	18	18	18	18
	1.5	2	4	4	6	6	6	8	8	10	10

demonstration we compute the Sharpe ratios of the four stock portfolios described in Section 3.2.2 over increasing horizons up to 20 years. Then we rank these portfolios by assigning rank 1 to the portfolio with the highest Sharpe ratio, rank 2 to the portfolio with the next highest Sharpe ratio, etc.

In the demonstration we employ the following four choices for the block length: 1 year, 5 years, block length equals the horizon length, and the “optimal” block length. In the first three cases we use the same block length for all return series. Since we use annual returns, the usage of 1-year blocks is equivalent to assuming that the returns of all stock portfolios are i.i.d. In the last case the block length is the “optimal” block length for each return series assuming the portfolio price process is given by the model presented in the preceding section. Since our primary purpose is a demonstration of consequences of using different block lengths, we do not perform a full-blown empirical estimation of the model parameters for each stock price process. Due to the absence of statistically significant evidence of mean reversion in the prices of portfolios of large and growth stocks, in the estimation of the mean and variance of long-term returns of these portfolios we employ the standard bootstrap method (this is equivalent to assuming either $\frac{\sigma_e}{\sigma_w} \rightarrow \infty$ or $\phi \rightarrow 1$). Since we have a strong statistical evidence of mean reversion in the prices of portfolios of small and value stocks, for both of these return series we assume the following model parameters: $\phi = 0.9$ and $\frac{\sigma_e}{\sigma_w} = 0.5$ (that are motivated by the discussions and results presented in the papers by Fama and French (1988) and Poterba and Summers (1988)). The optimal block length in this case can be found in Table 4 (when $n = 63$). In order to compute the Sharpe ratio of each portfolio for each investment horizon, we perform 10,000 block bootstrap simulations of historical returns. Then we estimate the mean return¹ and variance of returns. The value of the annualized risk-free interest rate is assumed to be 3%.

¹Note that the estimate for the mean return in our model is not biased for any block length.

The results of the simulations,² computations of the Sharpe ratios, and ranking of stock portfolios are reported in Table 5. Observe that when the returns are assumed to be i.i.d., for relatively short horizons the best portfolio is the value stocks, but as the horizon increases the rank of this portfolio decreases. For relatively long horizons the best portfolio is the large stocks, and the second best portfolio is the growth stocks. The small stock portfolio has always the lowest rank regardless of the horizon length. To understand the behavior of the Sharpe ratio in this case, we refer the interested readers to Levy (1972). In particular, Levy was the first to show that if the returns are i.i.d., while both the expected return and standard deviation increase with horizon, the rate of increase is greater for the standard deviation. Moreover, the rate of increase in standard deviation is much greater for portfolios with high mean returns and variance of returns. As a result, over long horizons the Sharpe ratios of portfolios with low standard deviations eventually become greater than the Sharpe ratios of portfolios with high standard deviations even though for short horizons the situation was directly opposite.

The use of 5-year blocks allows some effects of mean reversion in the prices of portfolios of small and value stocks to be captured. Still, for horizons longer than 5 years the risk of small and value stocks are strongly overestimated because, recall, the use of shorter blocks destroys serial dependence in returns. The best portfolio now is the value stocks regardless of the horizon length, the next best is the large stocks, and the worst is again the small stocks. The use of the block length that equals the horizon length produces close to optimal estimates for the standard deviations of returns on portfolios of small and value stocks, but at the same time the standard deviations of the returns on large and growth stocks are highly underestimated. This is because the use of long blocks produces a downward biased estimate of variance when the returns are i.i.d. Yet now the small stock portfolio increases its ranking from the worst to the second best as the investment horizon lengthens. Finally, when the block length is optimal for each portfolio, the small stock portfolio is the second best choice for horizons longer than 2 years, whereas the growth stock portfolio is the worst choice for the investor. To summarize, in our example the different choices for the block length lead to different values of the Sharpe ratios and different ranking of stock portfolios. The use of the moving block bootstrap method with the optimal block length for each stock portfolio produces virtually opposite ranking to that obtained using the standard bootstrap method.

6. Summary, Conclusions, and Recommendations

It is generally accepted that time horizon plays a crucial role in the determination of the optimal portfolio choice. Financial advisors and mutual fund sales literature commonly recommend that the investment horizon should be rather long in order to benefit from the “time diversification”. In finance, the problem of estimating risk and reward over a long horizon comes from lack of sufficient data and the uncertainty in the nature of the return generating process. To overcome these obstacles researchers rely heavily on statistical bootstrap methods. The main goal of this paper is to demonstrate potentially very serious pitfalls that can result from an improper application of a bootstrap method, and how to avoid these pitfalls.

Specifically, we considered a long-term investor whose goal was to rank the following four portfolios: small stocks, large stocks, growth stocks, and value stocks. First of all, we performed the

²The values in the table are based on the results of the simulations using the moving block bootstrap method. The use of the stationary bootstrap method for the case with the “optimal” block length (for the returns on the small and value stocks) produces virtually identical results. Similarly, instead of using the “optimal” block length in a block bootstrap method one can employ the standard bias correction method described in Subsection 4.3. The magnitudes of the bias in the estimation of the variance ratio for the case where $l = m$ can be found in Table 3 (when $n = 63$). This bias correction method also produces results close to the results reported in Table 5 Panel D.

Table 5. Sharpe ratios and the ranks (in brackets) of stock portfolios for different horizons and under different choices for the block length in the moving block bootstrap method. The “optimal” block length means using 1-year blocks (this is equivalent to the application of the standard bootstrap) for the returns on the large and growth stocks, and the optimal block length for the returns on the small and value stocks assuming the following model parameters: $\phi = 0.9$ and $\frac{\sigma_e}{\sigma_w} = 0.5$.

Horizon, years	Portfolio			
	Small stocks	Large stocks	Growth stocks	Value stocks
Panel A: Block length equals 1 year				
2	0.60 (4)	0.69 (2)	0.62 (3)	0.81 (1)
4	0.75 (4)	0.90 (2)	0.80 (3)	1.01 (1)
6	0.80 (4)	1.00 (2)	0.90 (3)	1.09 (1)
8	0.83 (4)	1.07 (2)	0.95 (3)	1.11 (1)
10	0.82 (4)	1.09 (2)	0.98 (3)	1.11 (1)
12	0.80 (4)	1.10 (1)	1.00 (3)	1.09 (2)
14	0.77 (4)	1.09 (1)	0.99 (3)	1.04 (2)
16	0.75 (4)	1.07 (1)	0.98 (3)	1.02 (2)
18	0.72 (4)	1.06 (1)	0.97 (2)	0.96 (3)
20	0.66 (4)	1.04 (1)	0.94 (2)	0.92 (3)
Panel B: Block length equals 5 years				
2	0.61 (4)	0.68 (2)	0.64 (3)	0.83 (1)
4	0.92 (4)	1.04 (2)	0.93 (3)	1.37 (1)
6	1.00 (4)	1.16 (2)	1.06 (3)	1.57 (1)
8	1.04 (4)	1.23 (2)	1.12 (3)	1.62 (1)
10	1.07 (4)	1.26 (2)	1.17 (3)	1.67 (1)
12	1.03 (4)	1.27 (2)	1.17 (3)	1.69 (1)
14	1.00 (4)	1.26 (2)	1.17 (3)	1.67 (1)
16	0.97 (4)	1.25 (2)	1.16 (3)	1.63 (1)
18	0.93 (4)	1.23 (2)	1.14 (3)	1.59 (1)
20	0.91 (4)	1.21 (2)	1.13 (3)	1.61 (1)
Panel C: Block length equals horizon length				
2	0.62 (4)	0.71 (2)	0.65 (3)	0.87 (1)
4	0.91 (4)	1.05 (2)	0.95 (3)	1.61 (1)
6	1.04 (4)	1.19 (2)	1.11 (3)	1.74 (1)
8	1.18 (4)	1.29 (2)	1.21 (3)	1.86 (1)
10	1.31 (3)	1.40 (2)	1.29 (4)	2.01 (1)
12	1.46 (2)	1.45 (3)	1.34 (4)	2.06 (1)
14	1.69 (2)	1.47 (3)	1.38 (4)	2.26 (1)
16	1.80 (2)	1.43 (3)	1.40 (4)	2.39 (1)
18	1.86 (2)	1.40 (3)	1.39 (4)	2.33 (1)
20	1.67 (2)	1.39 (3)	1.36 (4)	2.39 (1)
Panel D: “Optimal” block length				
2	0.62 (3)	0.69 (2)	0.62 (4)	0.86 (1)
4	0.96 (2)	0.89 (3)	0.80 (4)	1.60 (1)
6	1.05 (2)	1.00 (3)	0.90 (4)	1.73 (1)
8	1.12 (2)	1.06 (3)	0.95 (4)	1.81 (1)
10	1.15 (2)	1.09 (3)	0.98 (4)	1.86 (1)
12	1.18 (2)	1.10 (3)	1.00 (4)	1.88 (1)
14	1.20 (2)	1.10 (3)	0.99 (4)	1.94 (1)
16	1.21 (2)	1.09 (3)	0.98 (4)	1.96 (1)
18	1.24 (2)	1.07 (3)	0.95 (4)	1.99 (1)
20	1.27 (2)	1.05 (3)	0.93 (4)	1.96 (1)

statistical randomization tests where the null hypothesis was that the stock returns are independent over time. We found a strong statistical evidence of mean reverting behavior in the prices of small and value stocks. Consequently, a moving block bootstrap method is necessary in order to estimate correctly the long-run mean return and variance of returns for these portfolios. Using a well-established model for the equity price process where the price is composed of a permanent and a transitory component, we demonstrated the estimation bias in a moving block bootstrap

method and proposed how this bias can be mitigated. One of the bias reduction methods consists in choosing a proper block length. We found that the optimal block length depends mainly on the investment horizon and the ratio of the standard deviation of the permanent component in the stock prices to the standard deviation of the temporary component. In particular, we found that when the value of this ratio is rather small, then the optimal block length is close to the horizon length in the original moving block bootstrap and longer than the horizon length in the stationary bootstrap. In other cases, where the value of this ratio is rather large, the optimal block length is shorter than the horizon length in both of these methods. Finally we demonstrated that different choices for the block length in the moving block bootstrap method generally lead to different ranking of stock portfolios. We showed that an improper use of a moving block bootstrap method usually causes underestimation of the risk of a portfolio whose returns are independent over time and overestimation of the risk of a portfolio whose returns are mean-reverting.

All in all, we conclude that bootstrap methods are of great practical importance in finance and merit further extensive research. One should be very cautious, however, while applying the moving block bootstrap methods because they provide generally biased estimates. The correct accounting for the bias seems to be impossible without making some specific assumptions about the nature of the return generating process. In other words, our results imply that the choice of the optimal block length cannot be made arbitrarily and, generally, the optimal block length is model dependent. Yet our findings suggest that the use of longer blocks produces more correct results than the use of shorter blocks. In particular, we found that for sufficiently long horizons (longer than 12 years) the use of the block length that equals to the horizon length produces the same ranking of stock portfolios as with the use of the optimal block length. Therefore, we would recommend to implement a block bootstrap method with longer blocks rather than with shorter blocks. Moreover, we would recommend to implement the stationary bootstrap (proposed by Politis and Romano (1994)) instead of the more popular original moving block bootstrap (proposed by Künsch (1989) and further elaborated by Carlstein *et al.* (1998) and Graflund (2001)). This is because the estimates provided by the stationary bootstrap are much less variable and generally less biased as compared with those provided by the original moving block bootstrap.

Appendix: Derivation of the Formula for $VR(m)$

Note that z_t can be written as $MA(\infty)$

$$z_t = w_t + \sum_{i=1}^{\infty} \phi^i w_{t-i}.$$

This allows us to compute the variance of z_t

$$Var(z_t) = (1 + \phi^2 + \phi^4 + \phi^6 + \dots) \sigma_w^2 = \frac{1}{1 - \phi^2} \sigma_w^2.$$

The variance ratio is given by

$$VR(m) = \frac{Var(p_t - p_{t-m})}{m Var(p_t - p_{t-1})} = \frac{Var(\sum_{i=1}^m e_{t-m+i}) + Var(z_t - z_{t-m})}{m (Var(e_t) + Var(z_t - z_{t-1}))},$$

where

$$p_t - p_{t-1} = \mu + e_t + (1 - \phi)z_{t-1} + w_t.$$

Let us find $Var(z_t - z_{t-m}) = 2Var(z_t) - 2Cov(z_t, z_{t-m})$. We know that

$$\begin{aligned} z_t &= w_t + \phi w_{t-1} + \phi^2 w_{t-2} + \phi^3 w_{t-3} + \dots \\ &+ \phi^m w_{t-m} + \phi^{m+1} w_{t-m-1} + \phi^{m+2} w_{t-m-2} + \phi^{m+3} w_{t-m-3} + \dots, \\ z_{t-m} &= w_{t-m} + \phi w_{t-m-1} + \phi^2 w_{t-m-2} + \phi^3 w_{t-m-3} + \dots \end{aligned}$$

Therefore

$$Cov(z_t, z_{t-m}) = \phi^m (1 + \phi^2 + \phi^4 + \phi^6 + \dots) \sigma_w^2 = \frac{\phi^m}{1 - \phi^2} \sigma_w^2.$$

Thus

$$Var(z_t - z_{t-m}) = 2 \frac{1}{1 - \phi^2} \sigma_w^2 - 2 \frac{\phi^m}{1 - \phi^2} \sigma_w^2 = 2 \frac{1 - \phi^m}{1 - \phi^2} \sigma_w^2.$$

Observe that formula above is valid for any $m \geq 1$. Therefore, the formula for the variance ratio becomes

$$VR(m) = \frac{m\sigma_e^2 + 2 \frac{1 - \phi^m}{1 - \phi^2} \sigma_w^2}{m\sigma_e^2 + 2m \frac{1 - \phi}{1 - \phi^2} \sigma_w^2}.$$

The division of both the numerator and the denominator in the formula above by $\frac{\sigma_w^2}{1 - \phi^2}$ produces formula (5).

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